Norm
(Def: N0-N4)

Normed vector space
d(x,y) = ||x-y||

Metric/Distance
form norm

Metric Space
(Def: M0-M5)

Discrete d_0

d_p, d_infinite

Isometry
(Bijective Function)

Continuous function
(epsilon-delta)

Composition of
Continuous function

Function space

Sequence space

Abs. con.

Open ball

union of
open balls
= open set

Interior point

Open (set)

open depends on X

Continuous function
(Open ball)

Closed
Boundary
(closure\interior)

intersection of
closure of W, X\W

Proposition 1.5.18
Proposition 1.5.19
Proposition 1.5.22

pre-image
Continuous function
(Topological)

image
Open function
(Topological)

pre-image is
discrete, function is
continuous

Cartesian product

Lipschitz
equivalent metric

Lip. eq. metric
space(function)

one way

Topological
equivalent metric

Id: d,d'

Homeomorphic
bij. cts. cts.

all d on R^n
different open balls
same open sets

homeomorphic intervals

Closed ball

pre-image
Continuous function
(Topological)

Nowhere dense
(complement of closure dense)
(lemma 1.8.25)
(Proposition 1.8.26)

Dense

Closure

Bounded
subset

Diameter

W, f(W)
bounded

finite union bounded

Bounded
function

Closure W = W & Acc(W)
Acc = lim w_n

Acc(W)

Iso(W) = W\Acc(W)

Closed set

singleton {x} closed

complement
Proposition 1.8.10

Proposition 1.8.9
Proposition 1.8.12
Proposition 1.8.15

Sequence

Continuous function
(Sequence)

Closure
seq. criterion

Open/Closed
seq. criterion